

Fitted Q-Evaluation with a Random Horizon Length

1 Problem Formulation

Consider an episodic Markov decision process (MDP) $M = (\mathcal{S}, \mathcal{A}, P, r, H)$, where H is the given horizon length. The goal of off-policy evaluation is to estimate the expected cumulative return (i.e., value) of a set of target policies $\pi = \{\pi_1, \dots, \pi_H\}$ from a fixed initial distribution ξ_1 based on a given dataset. $\pi_h(a|s) = P(a_h = a|s_h = s)$ represents the policy distribution at step h . However, in many practical scenarios, we only have access to a dataset where the horizon length of each trajectory is random. We would like to leverage this data to evaluate the policy under a given horizon length H . Here we formalize the model as follows.

Suppose that the batch data $\mathcal{D} = \{\mathcal{D}_1, \dots, \mathcal{D}_K\}$ consists of K independent episodes collected using an unknown behavior policy $\bar{\pi}$. Each episode $\mathcal{D}_k = \{(s_{h'}^k, a_{h'}^k, r_{h'}^k)\}_{h' \in [H_k]}$ is a trajectory of H_k state-transition tuples. $H_1, \dots, H_K$ are *i.i.d.* random variables. Let $N = \sum_{k=1}^K H_k$ be the total number of sample transitions. We aim to evaluate the Q-function:

$$Q_h(s, a) = \mathbb{E} \left[\sum_{h'=h}^H r(s_{h'}, a_{h'}) \middle| s_h = s, a_h = a \right],$$

and the expected cumulative return from a fixed initial distribution ξ_1 :

$$v_\pi = \mathbb{E} \left[\sum_{h=1}^H r(s_h, a_h) \middle| s_1 \sim \xi_1, a_h \sim \pi_h(\cdot | s_h) \right].$$

The Q-function satisfies the Bellman equation for policy $\pi = \{\pi_1, \dots, \pi_H\}$

$$Q_h(s, a) = r(s, a) + \mathbb{E}[V_{h+1}(s') | s_h = s, a_h = a] \quad (1)$$

$$V_{h+1}(s') = \int_{a' \in \mathcal{A}} Q_{h+1}(s', a') \pi_{h+1}(a' | s') da' \quad (2)$$

Throughout the paper, we apply Fitted Q-Evaluation (FQE) to address the problem. Fitted Q-Evaluation is a variant of the Fitted Q-Iteration method. Let $\mathcal{F}$ be a given linear function class spanned by φ . We initialize by setting $\hat{Q}_{H+1} = 0$. For $h = H, \dots, 1$, we recursively estimate Q_h through regression and function approximation:

$$\hat{Q}_h = \operatorname{argmin}_{f \in \mathcal{F}} \left\{ \frac{1}{\sum_{k=1}^K H_k} \sum_{k=1}^K \sum_{h'=1}^{H_k} (f(s_{h'}^k, a_{h'}^k) - y_{hh'}^k)^2 + \frac{1}{\sum_{k=1}^K H_k} \lambda \rho(f) \right\} \quad (3)$$

where

$$y_{hh'}^k = r_{h'}^k + \int_a \hat{Q}_{h+1}(s_{h'+1}^k, a) \pi_{h+1}(a | s_{h'+1}^k) da.$$

and $\rho(f)$ is a proper regularizer. The estimated return is

$$\hat{v}_\pi = \int_s \int_a \hat{Q}_1(s, a) \xi_1(s) \pi_1(a|s) ds da.$$

Assumption 1. Let $H_1, \dots, H_K$ be independent random variables for all $k \in [K]$, each sub-exponential. We denote that $\bar{H} := \mathbb{E}(H_k)$, and the sub-exponential norm $\|H_k\|_{\psi_1}$.

Assumption 2 (Policy Completeness). For any $f \in \mathcal{F}$, $\mathcal{P}(f) \in \mathcal{F}$, $r \in \mathcal{F}$.

Assumption 3. FQE is conducted under linear function approximation, i.e. $\mathcal{F}$ is a linear function class spanned by φ , $\|\varphi(s, a)\|_\infty \leq 1$.

1.0.1 Notation

Here are some notations for the following analysis.

$$\Sigma = \mathbb{E} \left[\frac{1}{N} \sum_{k=1}^K \sum_{h'=1}^{H_k} \varphi(s_{h'}^k, a_{h'}^k) \varphi(s_{h'}^k, a_{h'}^k)^\top \right] \quad (4)$$

$$\varphi^\pi(s) := \int \varphi(s, a) \pi(a|s) da \quad (5)$$

$$\varphi(s, a)^\top M^\pi = \mathbb{E}[\varphi^\pi(s')^\top | s, a], M^\pi \text{ is the matrix embedding of } \mathcal{P}^\pi \quad (6)$$

$$\nu_1 = \mathbb{E}[\varphi(s, a) | s \sim \xi_1, a \sim \pi(\cdot|s)] \quad (7)$$

$$\nu_h = \mathbb{E}[\varphi(s_h, a_h) | s_1 \sim \xi_1, a_h \sim \pi_h(\cdot|s_h)] \quad (8)$$

We further extend the notations in (5) and (6) such that

$$\varphi_h(s) := \int \varphi(s, a) \pi_h(a|s) da = \int \varphi(s, a) P(a_h = a | s_h = s) da \quad (9)$$

$$\varphi(s, a)^\top M_h = \mathbb{E}[\varphi_h(s')^\top | s, a] \quad (10)$$

(6) implies that

$$\varphi^\pi(s)^\top M^\pi = \mathbb{E}[\varphi^\pi(s')^\top | s' \sim p^\pi(\cdot|s)], p^\pi(s'|s) = \int_{\mathcal{A}} \pi(a|s) p(s'|s, a) da \quad (11)$$

and we can infer from (8) that

$$\nu_h = M_h^\top M_{h-1}^\top \dots M_2^\top \nu_1 \quad (12)$$

Denote

$$\mathcal{P}^\pi(f) = \mathbb{E}[f(s', a') | s, a] = \int_{s' \in \mathcal{S}} \int_{a' \in \mathcal{A}} f(s', a') \pi(a'|s') p(s'|s, a) ds' da' \quad (13)$$

Then the Bellman Equation can be written as

$$Q_h(s, a) = r(s, a) + \mathcal{P}^{\pi_{h+1}}(Q_{h+1}^\pi)(s, a) := r(s, a) + \mathcal{P}_{h+1}(Q_{h+1}^\pi)(s, a) \quad (14)$$

2 Asymptotic Normality of FQE

Theorem 2.1 (Asymptotic Normality of FQE). *Denote*

$$\Sigma_0 = \mathbb{E} \left[\frac{1}{H} \sum_{h'=1}^{H_1} \varphi(s_{h'}^1, a_{h'}^1) \varphi(s_{h'}^1, a_{h'}^1)^\top \right]. \quad (15)$$

Suppose $\lambda_{\min}(\Sigma_0) > 0$, and Assumption 2 holds. The FQE with linear function approximation is asymptotically normal:

$$\sqrt{N} (\hat{v}_\pi - v_\pi) \xrightarrow{d} \mathcal{N}(0, \sigma^2) \text{ as } K \rightarrow \infty, N = \sum_{k=1}^K H_k$$

where $\xrightarrow{d}$ denotes converging in distribution. The asymptotic variance σ^2 is given by

$$\sigma^2 = \frac{1}{H} \sum_{h=1}^H \nu_h^\top \Sigma_0^{-1} \Omega_{h,h} \Sigma_0^{-1} \nu_h + \frac{2}{H} \sum_{h_1 < h_2}^H \nu_{h_1}^\top \Sigma_0^{-1} \Omega_{h_1, h_2} \Sigma_0^{-1} \nu_{h_2}$$

where $\nu_h = \mathbb{E}[\varphi(s_h, a_h) \mid s_1 \sim \xi_1, a_h \sim \pi_h(\cdot \mid s_h)]$, $\epsilon_{hh'}^k = Q_h(s_{h'}^k, a_{h'}^k) - r_{h'}^k - V_{h+1}(s_{h'+1}^k)$ and

$$\Omega_{h_1, h_2} := \mathbb{E} \left(\sum_{h'=1}^{H_1} \varphi(s_{h'}^1, a_{h'}^1) \varphi(s_{h'}^1, a_{h'}^1)^\top \epsilon_{h_1 h'}^1 \epsilon_{h_2 h'}^1 \right) \quad (16)$$

3 Estimation of σ^2

To estimate the asymptotic variance σ^2 by plugging in sample values, we need to first estimate the following quantities:

$$\begin{aligned} \epsilon_{hh'}^k &= Q_h^\pi(s_{h'}^k, a_{h'}^k) - r_{h'}^k - V_{h+1}^\pi(s_{h'+1}^k), \\ \nu_h &= \mathbb{E}[\varphi(s_h, a_h) \mid s_1 \sim \xi_1, a_h \sim \pi_h(\cdot \mid s_h)], \\ \Omega_{h_1, h_2} &= \mathbb{E} \left(\sum_{h'=1}^{H_1} \varphi(s_{h'}^1, a_{h'}^1) \varphi(s_{h'}^1, a_{h'}^1)^\top \epsilon_{h_1 h'}^1 \epsilon_{h_2 h'}^1 \right), \\ \Sigma_0 &= \mathbb{E} \left[\frac{1}{H} \sum_{h'=1}^{H_1} \varphi(s_{h'}^1, a_{h'}^1) \varphi(s_{h'}^1, a_{h'}^1)^\top \right]. \end{aligned}$$

The calculation proceeds as follows:

We first split the data into two batches, A_1 and A_2 . The first batch A_1 is used to estimate the function by fitted Q-iteration, i.e. Q_h and V_h .

We then evaluate the function values $\varphi(s_{h'}^k, a_{h'}^k)$, $Q_h(s_{h'+1}^k, a_{h'}^k)$, and $V_{h+1}(s_{h'+1}^k)$ by plugging in the data from A_2 . Using samples $(s_{h'+1}^k, s_{h'}^k, a_{h'}^k) \in A_2$, we obtain the value of $\epsilon_{hh'}^k$.

Next, we will estimate ν_h . From Equation (12), ν_h is essentially estimated through $M_h, h = 2, \dots, H$. Further notice that the definition of M_{h+1} in (10) is equivalent to

$$M_{h+1}^\top \varphi(s, a) = \mathbb{E}[\varphi(s_{h+1}, a_{h+1}) \mid s_h = s, a_h = a] = \mathcal{P}_{h+1} \varphi(s, a) \quad (17)$$

Apply the estimator of $\mathcal{P}\varphi$ in Equation (31) and using samples from A_2 , we have an estimation of $\mathcal{P}_{h+1}\varphi$, i.e.

$$\widehat{\mathcal{P}}_{h+1}\varphi(s, a) = \widehat{M}_{h+1, A_2}\varphi(s, a) \quad (18)$$

where

$$\widehat{M}_{h, A_2} = \widehat{\Sigma}_{A_2}^{-1} \sum_{(s_n, a_n, s'_n) \in A_2} \varphi(s_n, a_n) \varphi_h^\top(s'_n) \quad (19)$$

$$\widehat{\Sigma}_{A_2} = \sum_{(s_n, a_n, s'_n) \in A_2} \varphi(s_n, a_n) \varphi^\top(s_n, a_n) + \lambda I \quad (20)$$

Thus, $\widehat{M}_{h+1, A_2}$ is the estimator of M_{h+1} using sample from A_2 . This is equivalent to solving the following optimization problem:

$$\widehat{M}_{h+1, A_2} = \operatorname{argmin}_M \sum_{(s, a, s') \in A_2} \left\| \varphi(s, a)^\top M - \widehat{\mathcal{P}}_{h+1}\varphi(s, a) \right\|_2^2. \quad (21)$$

Compute $\nu_1 = \sum_{s \in \mathcal{S}, a \in A(s)} \xi_1(s) \varphi(s, a) \pi_1(a|s)$. Then, by (12),

$$\widehat{\nu}_{h, A_2}^\top = \nu_1^\top \widehat{M}_{2, A_2} \widehat{M}_{3, A_2} \cdots \widehat{M}_{h, A_2}.$$

We estimate Ω_{h_1, h_2} and Σ_0 by their sample means. For $(s_{h'+1}^k, s_{h'}^k, a_{h'}^k) \in A_2$:

$$\widehat{\Omega}_{h_1, h_2} = \frac{1}{K_{A_2}} \sum_{k=1}^{K_{A_2}} \sum_{h'=1}^{H_k} \varphi(s_{h'}^k, a_{h'}^k) \varphi(s_{h'+1}^k, a_{h'+1}^k)^\top \epsilon_{h_1 h'}^k \epsilon_{h_2 h'}^k, \quad (22)$$

$$\widehat{\Sigma}_0 = \frac{1}{K_{A_2} \overline{H}} \sum_{k=1}^{K_{A_2}} \sum_{h'=1}^{H_k} \varphi(s_{h'}^k, a_{h'}^k) \varphi(s_{h'}^k, a_{h'}^k)^\top + \lambda I. \quad (23)$$

for some small λ , where K_{A_2} is the number of trajectories in A_2 .

By swapping the datasets A_1 and A_2 —using A_2 for function estimation and A_1 for plugging in values—we obtain another estimate of σ^2 . Averaging the two estimates yields our final estimate of σ^2 .

3.1 Cliff-walking Simulation

3.1.1 Environment Setting

In this section, we conduct experiments in the cliff walking environment. The agent starts at the start state and attempts to navigate across the cliff to reach the goal state. Each step incurs a reward of -1 , except for those leading to cliff states, which result in a reward of -100 .

Our objective in this paper is to evaluate the target policy using a dataset with a random horizon length. To achieve this, we first generate a set of random variables $H_1, H_2, \dots, H_k$ to represent the horizon lengths. For each episode, we simulate transitions and terminate the episode once the number of transitions reaches the assigned horizon length.

There are three potential ways to terminate the process: the agent either reaches the cliff states, the goal state, or the step limit. In our setting, the first two termination conditions—the cliff states and the goal state—are treated as absorbing states, where the agent remains in the state thereafter, while the process ends once the step limit is reached. If the agent remains in a cliff state after falling or stays in the goal state after reaching it, and the process terminates only when the step limit is reached, the remaining transitions would be redundant, reducing the informativeness of the dataset.

To address this, we modify the transition dynamics so that the agent is immediately sent back to the start state upon reaching a cliff. This prevents unproductive sequences, increases the diversity of transitions, and ensures better use of the random horizon lengths. Due to these dynamics, the environment contains only 38 valid states, with the goal state being the only absorbing state.

We also modify the environment in order to make it stochastic.

3.1.2 Policy Construction

We first apply value iteration to compute the solution to the Bellman optimality equation:

$$Q_h(s, a) = \mathbb{E} \left[r + \max_{a'} Q_{h+1}(s', a') \mid s, a \right],$$

with $Q_{H+1}(s, a) = 0$. After obtaining $Q_h^{GT}(s, a)$ for each pair of (s, a) and h , an estimated deterministic optimal policy is constructed as:

$$\pi_h(s) = \arg \max_a Q_h^{GT}(s, a), \quad h = 1, \dots, H.$$

To introduce exploration, we construct an ϵ -greedy policy by adding probability ϵ of selecting a random action.

3.1.3 Simulation Results

In this simulation, we set the randomness of transitions to 0.05, 0.1, 0.2. The target policy is the greedy policy constructed as described in 3.1.2

The behavior policy is an ϵ -greedy policy, where we set ϵ to be relatively large: 0.7. This is because $H = 5$ is relatively small, so if the behavior ϵ is small, then the policy will be close to the one constructed in 3.1.2 with $H = 5$, which leads to low rank of Σ_0 . The increased exploration ensures that the trajectory generated by the behavior policy will cover more of the state space, thereby satisfying the assumption that $\lambda_{\min}(\Sigma_0) > 0$.

In Fig 1, the setting is as follows: $H = 20$, $\bar{H} = 18$. The subsample size is $K^{0.8}$. The number of bootstrap samples is 100, and we repeat the experiment 100 times.

A Proof of the Asymptotic Normality

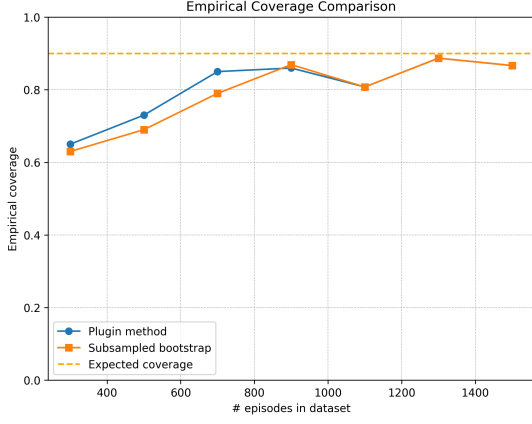
A.1 Decomposition of $\hat{v}_\pi - v_\pi$

First let us study $\hat{v}_\pi - v_\pi$ from its decomposition.

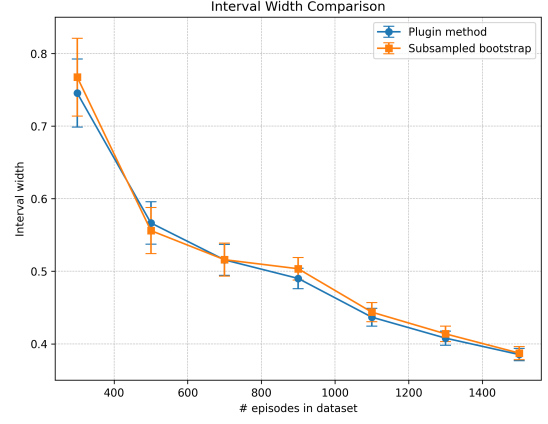
Under the linear approximation assumption, $\hat{Q}_h^\pi = \varphi(s, a)^\top \hat{w}_h$

If it is a linear approximation, (3) can be further written into

$$\hat{w}_h = \operatorname{argmin}_{w \in \mathbb{R}^d} \left\{ \frac{1}{\sum_{k=1}^K H_k} \sum_{k=1}^K \sum_{h'=1}^{H_k} (\varphi(s_{hh'}^k, a_{hh'}^k)^\top w - y_{hh'}^k)^2 + \frac{1}{\sum_{k=1}^K H_k} \lambda \|\omega\|_2^2 \right\} \quad (24)$$

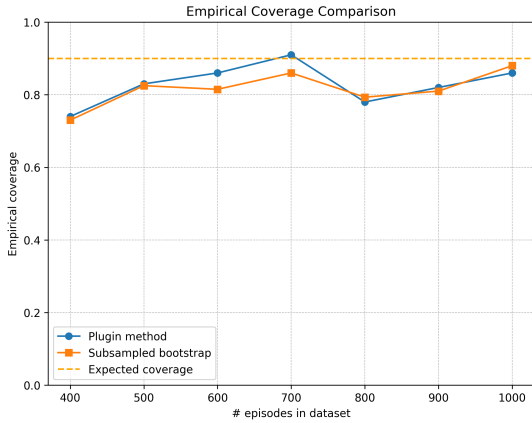


(a) Empirical coverage of CI, transition randomness 0.01

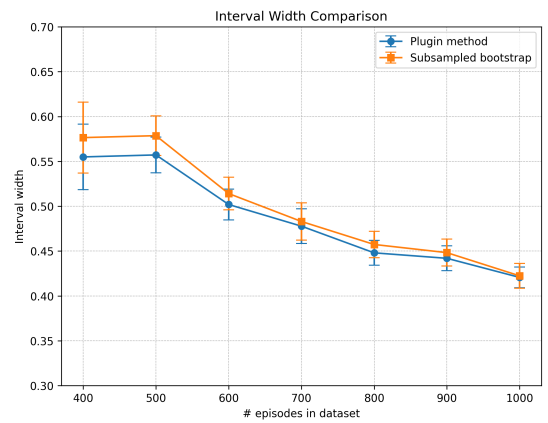


(b) Confidence interval width, transition randomness 0.01

Figure 1: Comparison of Empirical Coverage and Confidence Interval Width for $\bar{H} = 18, H = 20$. Randomness of transitions 0.01



(a) Empirical coverage of CI, transition randomness 0.01



(b) Confidence interval width, transition randomness 0.01

Figure 2: Comparison of Empirical Coverage and Confidence Interval Width for $\bar{H} = 18, H = 15$. Randomness of transitions 0.01.

Denote

$$\widehat{\Sigma} = \sum_{k=1}^K \sum_{h'=1}^{H_k} \varphi(s_{h'}^k, a_{h'}^k) \varphi(s_{h'}^k, a_{h'}^k)^\top + \lambda I \quad (25)$$

$$\widetilde{R} = \sum_{k=1}^K \sum_{h'=1}^{H_k} r_{h'}^k \varphi(s_{h'}^k, a_{h'}^k), \quad (26)$$

$$\widetilde{M}_h = \sum_{k=1}^K \sum_{h'=1}^{H_k} \varphi(s_{h'}^k, a_{h'}^k) \varphi_h(s_{h'+1}^k)^\top, \varphi_h(s) = \int_a \varphi(s, a) \pi_h(a|s) da \quad (27)$$

$$\widehat{R} = \widehat{\Sigma}^{-1} \widetilde{R}, \widehat{M}_h = \widehat{\Sigma}^{-1} \widetilde{M}_h. \quad (28)$$

We shall obtain from (24) that

$$\widehat{w}_h = \widehat{R} + \widehat{M}_{h+1} \widehat{w}_{h+1}$$

Furthermore, let

$$\mathcal{P}^\pi(f) = \mathbb{E}[f(s', a') | s, a] = \int_{s' \in \mathcal{S}} \int_{a' \in \mathcal{A}} f(s', a') \pi(a' | s') p(s' | s, a) ds' da'$$

Then the Bellman Equation can be written as

$$Q_h^\pi(s, a) = r(s, a) + \mathcal{P}^{\pi_{h+1}}(Q_{h+1}^\pi)(s, a) \quad (29)$$

Thus,

$$Q_1^\pi = (\mathcal{J} + \mathcal{P}^{\pi_2} + \mathcal{P}^{\pi_2} \mathcal{P}^{\pi_3} + \dots + \mathcal{P}^{\pi_2} \mathcal{P}^{\pi_3} \mathcal{P}^{\pi_4} \dots \mathcal{P}^{\pi_H}) r = \left(\mathcal{J} + \sum_{h=2}^H \prod_{k=2}^h \mathcal{P}^{\pi_k} \right) r \quad (30)$$

We estimate $\mathcal{P}^\pi(f)$ by

$$\widehat{\mathcal{P}}^\pi(f) = \operatorname{argmin}_{g \in \mathcal{F}} \left\{ \frac{1}{N} \sum_{k=1}^K \sum_{h'=1}^{H_k} \left(g(s_{h'}^k, a_{h'}^k) - \int_A f(s_{h'+1}^k, a) \pi(a | s_{h'+1}^k) da \right)^2 + \frac{\lambda}{N} \rho(g) \right\} \quad (31)$$

where $N = \sum_{k=1}^K H_k$.

For the case where $f = \widehat{Q}_{h+1}^\pi \in \mathcal{F}$, by Assumption 2, $\mathcal{P}^{\pi_{h+1}}(\widehat{Q}_{h+1}^\pi) \in \mathcal{F}$, $r \in \mathcal{F}$. Then we could apply ridge regression to estimate them as well.

$$\widehat{w}_h^1 = \operatorname{argmin}_{w \in \mathbb{R}^d} \left\{ \frac{1}{N} \sum_{k=1}^K \sum_{h'=1}^{H_k} \left(\varphi(s_{h'}^k, a_{h'}^k)^\top w - \int_A \varphi(s_{h'+1}^k, a) \widehat{w}_{h+1} \pi_{h+1}(a | s_{h'+1}^k) da \right)^2 + \frac{\lambda}{N} \|w\|^2 \right\}$$

That is, $\widehat{w}_h^1 = \widehat{M}_{h+1} \widehat{w}_{h+1}$. Therefore, $\widehat{\mathcal{P}}^{\pi_{h+1}}(\widehat{Q}_{h+1}^\pi) = \varphi(s, a)^\top \widehat{w}_h^1 = \varphi(s, a)^\top \widehat{M}_{h+1} \widehat{w}_{h+1}$.

Similarly, let

$$\widehat{r} = \operatorname{argmin}_{h \in \mathcal{F}} \left\{ \frac{1}{N} \sum_{k=1}^K \sum_{h'=1}^{H_k} (h(s_{h'}^k, a_{h'}^k) - r_{h'}^k)^2 + \frac{\lambda}{N} \rho(h) \right\}$$

It is easy to get that $\hat{r} = \varphi(s, a)^\top \hat{R}$.

Combining the results above,

$$\hat{Q}_h^\pi = \hat{\mathcal{P}}^{\pi_{h+1}}(\hat{Q}_{h+1}^\pi) + \hat{r} = \varphi(s, a)^\top \left(\widehat{M}_{h+1} \widehat{w}_{h+1} + \widehat{R} \right) \quad (32)$$

Applying this relation and the Bellman equation recursively,

$$\hat{Q}_1^\pi = \left(\mathcal{I} + \hat{\mathcal{P}}^{\pi_2} + \hat{\mathcal{P}}^{\pi_2} \hat{\mathcal{P}}^{\pi_3} + \dots + \hat{\mathcal{P}}^{\pi_2} \hat{\mathcal{P}}^{\pi_3} \hat{\mathcal{P}}^{\pi_4} \dots \hat{\mathcal{P}}^{\pi_H} \right) \hat{r} = \left(\mathcal{I} + \sum_{h=2}^H \prod_{k=2}^h \hat{\mathcal{P}}^{\pi_k} \right) \hat{r} \quad (33)$$

By Lemma (?), we have

$$Q_1^\pi - \hat{Q}_1^\pi = \sum_{h=1}^H \left(\prod_{k=2}^h \hat{\mathcal{P}}_k \right) \left(Q_h^\pi - \hat{\mathcal{P}}_{h+1} Q_{h+1}^\pi - \hat{r} \right)$$

and note that

$$\begin{aligned} Q_h^\pi - \hat{\mathcal{P}}_{h+1} Q_{h+1}^\pi - \hat{r} &= \varphi(s, a)^\top \widehat{\Sigma}^{-1} \left(\widehat{\Sigma} w_h - \sum_{k=1}^K \sum_{h'=1}^{H_k} \varphi(s_{h'}^k, a_{h'}^k) (r_{h'}^k + \varphi^{\pi_{h+1}}(s_{h'+1}^k)^\top w_{h+1}) \right) \\ &= \lambda \varphi(s, a)^\top \widehat{\Sigma}^{-1} w_h \\ &\quad + \sum_{k=1}^K \sum_{h'=1}^{H_k} \varphi(s, a)^\top \widehat{\Sigma}^{-1} \varphi(s_{h'}^k, a_{h'}^k) (Q_h^\pi(s_{h'}^k, a_{h'}^k) - r_{h'}^k - V_{h+1}^\pi(s_{h'+1}^k)) \end{aligned}$$

where $V_h^\pi(s) = \int_a \widehat{Q}_h^\pi(s, a) \pi_h(a|s) da$.

Therefore,

$$\begin{aligned} Q_1^\pi - \hat{Q}_1^\pi &= \sum_{h=1}^H \left(\lambda \varphi(s, a)^\top \prod_{k=2}^h \widehat{M}_k \widehat{\Sigma}^{-1} w_h \right) \\ &\quad + \sum_{h=1}^H \left(\sum_{k=1}^K \sum_{h'=1}^{H_k} \varphi(s, a)^\top \prod_{k=2}^h \widehat{M}_k \widehat{\Sigma}^{-1} \varphi(s_{h'}^k, a_{h'}^k) (Q_h^\pi(s_{h'}^k, a_{h'}^k) - r_{h'}^k - V_{h+1}^\pi(s_{h'+1}^k)) \right) \end{aligned}$$

Consequently, recall the notation

$$(\widehat{\nu}_h^\pi)^\top = (\nu_1^\pi)^\top \prod_{k=2}^h \widehat{M}_k, \quad \nu_1^\pi = \mathbb{E}_{s \sim \xi_1, a \sim \pi_1} \varphi(s, a)$$

then

$$\begin{aligned} v_\pi - \widehat{v}_\pi &= \mathbb{E}_{s \sim \xi_1, a \sim \pi(\cdot|s)} \left[Q_1^\pi - \widehat{Q}_1^\pi \right] \\ &= \sum_{h=1}^H \left(\lambda (\widehat{\nu}_h^\pi)^\top \widehat{\Sigma}^{-1} w_h + \sum_{k=1}^K \sum_{h'=1}^{H_k} (\widehat{\nu}_h^\pi)^\top \widehat{\Sigma}^{-1} \varphi(s_{h'}^k, a_{h'}^k) (Q_h^\pi(s_{h'}^k, a_{h'}^k) - r_{h'}^k - V_{h+1}^\pi(s_{h'+1}^k)) \right) \end{aligned}$$

Recall the definition

$$\Sigma = \mathbb{E} \left[\frac{1}{N} \sum_{k=1}^K \sum_{h'=1}^{H_k} \varphi(s_{h'}^k, a_{h'}^k) \varphi(s_{h'}^k, a_{h'}^k)^\top \right]. \quad (34)$$

where $N = \sum_{k=1}^K H_k$.

Thus,

$$\sqrt{N}(v_\pi - \hat{v}_\pi) = E_1 + E_2 + E_3$$

where

$$\begin{aligned} E_1 &= \frac{1}{\sqrt{N}} \sum_{h=1}^H \sum_{k=1}^K \sum_{h'=1}^{H_k} (\nu_h^\pi)^\top \Sigma^{-1} \varphi(s_{h'}^k, a_{h'}^k) (Q_h^\pi(s_{h'}^k, a_{h'}^k) - r_{h'}^k - V_{h+1}^\pi(s_{h'+1}^k)) \\ E_2 &= \sum_{h=1}^H \left(N (\hat{\nu}_h^\pi)^\top \hat{\Sigma}^{-1} - (\nu_h^\pi)^\top \Sigma^{-1} \right) \left(\frac{1}{\sqrt{N}} \sum_{k=1}^K \sum_{h'=1}^{H_k} \varphi(s_{h'}^k, a_{h'}^k) (Q_h^\pi(s_{h'}^k, a_{h'}^k) - r_{h'}^k - V_{h+1}^\pi(s_{h'+1}^k)) \right) \\ E_3 &= \sqrt{N} \sum_{h=1}^H \lambda (\hat{\nu}_h^\pi)^\top \hat{\Sigma}^{-1} w_h \end{aligned}$$

A.2 Asymptotic Distribution of E_1

Recall that

$$\Omega_{h_1, h_2} := \mathbb{E} \left(\sum_{h'=1}^{H_1} \varphi(s_{h'}^1, a_{h'}^1) \varphi(s_{h'}^1, a_{h'}^1)^\top \epsilon_{h_1 h'}^1 \epsilon_{h_2 h'}^1 \right)$$

Further assuming that Σ^{-1} is not ill-conditioned, i.e. eigenvalue of Σ^{-1} is upper-bounded by some constant C_1 , that is,

Assumption 4. $\varphi(s, a)^\top \Sigma^{-1} \varphi(s, a) \leq C_1 d$.

Notice that this assumption can also be viewed as an assumption on Σ_0 , due to the convergence that $\Sigma \rightarrow \Sigma_0$ by Lemma A.3.

Theorem A.1. $E_1 \xrightarrow{d} \mathcal{N}(0, \sigma_1^2 + 2\sigma_2^2)$, where

$$\sigma_1^2 = \frac{1}{H} \sum_{h=1}^H (\nu_h^\pi)^\top \Sigma^{-1} \Omega_{h, h} \Sigma^{-1} \nu_h^\pi \quad (35)$$

$$\sigma_2^2 = \frac{1}{H} \sum_{h_1 < h_2}^H (\nu_{h_1}^\pi)^\top \Sigma^{-1} \Omega_{h_1, h_2} \Sigma^{-1} \nu_{h_2}^\pi \quad (36)$$

Proof. Denote

$$e_{h'}^k = \frac{1}{\sqrt{K}} \sum_{h=1}^H (\nu_h^\pi)^\top \Sigma^{-1} \varphi(s_{h'}^k, a_{h'}^k) (Q_h^\pi(s_{h'}^k, a_{h'}^k) - r_{h'}^k - V_{h+1}^\pi(s_{h'+1}^k))$$

and thus,

$$E_1 = \frac{1}{\sqrt{H_k}} \sum_{k=1}^K \sum_{h'=1}^{H_k} e_{h'}^k,$$

where $\bar{H}_k = \sum_{k=1}^K H_k / K$.

Given $H_1, \dots, H_K$, let the filtration to be $\mathcal{F}_{h', k}, k = 1, \dots, K$ generated by $(s_1^1, a_1^1, s_2^1), \dots, (s_{H_1-1}^1, a_{H_1-1}^1, s_{H_1}^1), (s_{H_1}^1, a_{H_1}^1), \dots, (s_1^k, a_1^k, s_2^k), \dots, (s_{h'}^k, a_{h'}^k)$, then

$$\mathbb{E}(e_{h'}^k | \mathcal{F}_{h', k}) = 0$$

Therefore, $\{e_{h'}^k\}_{h' \in [H_k]}^{k \in [K]}$ is a martingale difference sequence. To show the asymptotic normality, we use the following martingale central limit theorem.

Theorem A.2. Let $\{X_{h'k}; h' \in [H_k], k \in [K]\}$ be a martingale difference array (row-wise) on the probability triple $(\Omega, \mathcal{F}, P)$, i.e. $\mathbb{E}(X_{h'k} | \mathcal{F}_{h',k}) = 0$. Suppose $X_{h'k}$ satisfy the following two conditions:

$$\max_{1 \leq h' \leq H_k, 1 \leq k \leq K} |X_{h'k}| \xrightarrow{p} 0, \text{ and } \sum_{k=1}^K \sum_{h'=1}^{H_k} X_{h'k}^2 \xrightarrow{p} \sigma^2$$

for $K \rightarrow \infty$. Also, $\max_{1 \leq h' \leq H_k, 1 \leq k \leq K} |X_{h'k}|$ is uniformly bounded in L_2 norm. Then $\sum_{k=1}^K \sum_{h'=1}^{H_k} X_{h'k} \xrightarrow{d} \mathcal{N}(0, \sigma^2)$ as $K \rightarrow \infty$.

To apply the martingale central limit theorem, we need to verify the following two conditions.

$$\max_{1 \leq h' \leq H_k, 1 \leq k \leq K} \left| \frac{1}{\sqrt{K}} \sum_{h=1}^H (\nu_h^\pi)^\top \Sigma^{-1} \varphi(s_{h'}^k, a_{h'}^k) (Q_h^\pi(s_{h'}^k, a_{h'}^k) - r_{h'}^k - V_{h+1}^\pi(s_{h'+1}^k)) \right| \xrightarrow{p} 0 \quad (37)$$

$$\sum_{k=1}^K \sum_{h'=1}^{H_k} \left[\frac{1}{\sqrt{K}} \sum_{h=1}^H (\nu_h^\pi)^\top \Sigma^{-1} \varphi(s_{h'}^k, a_{h'}^k) (Q_h^\pi(s_{h'}^k, a_{h'}^k) - r_{h'}^k - V_{h+1}^\pi(s_{h'+1}^k)) \right]^2 \xrightarrow{p} \sigma^2 \quad (38)$$

To check condition (37), notice that

$$\left| Q_h^\pi(s_{h'}^k, a_{h'}^k) - r_{h'}^k - V_{h+1}^\pi(s_{h'+1}^k) \right| \leq H - h + 1.$$

Also,

$$\begin{aligned} & \max_{1 \leq h' \leq H_k, 1 \leq k \leq K} \left| \frac{1}{\sqrt{K}} \sum_{h=1}^H (\nu_h^\pi)^\top \Sigma^{-1} \varphi(s_{h'}^k, a_{h'}^k) (Q_h^\pi(s_{h'}^k, a_{h'}^k) - r_{h'}^k - V_{h+1}^\pi(s_{h'+1}^k)) \right| \quad (39) \\ & \leq \max_{1 \leq h' \leq H_k, 1 \leq k \leq K} \frac{1}{\sqrt{K}} \sum_{h=1}^H (H - h + 1) \left| (\nu_h^\pi)^\top \Sigma^{-1} \varphi(s_{h'}^k, a_{h'}^k) \right| \rightarrow 0, \end{aligned}$$

as $\sum_{h=1}^H (H - h + 1) \left| (\nu_h^\pi)^\top \Sigma^{-1} \varphi(s_{h'}^k, a_{h'}^k) \right|$ is independent of $H_1, \dots, H_K$ and K . We have verified condition (37).

For condition (38), let $\epsilon_{hh'}^k = Q_h^\pi(s_{h'}^k, a_{h'}^k) - r_{h'}^k - V_{h+1}^\pi(s_{h'+1}^k)$,

$$\begin{aligned} I_1 &:= \sum_{h=1}^H (\nu_h^\pi)^\top \Sigma^{-1} \left(\frac{1}{K} \sum_{k=1}^K \sum_{h'=1}^{H_k} \varphi(s_{h'}^k, a_{h'}^k) \varphi(s_{h'}^k, a_{h'}^k)^\top (\epsilon_{hh'}^k)^2 \right) \Sigma^{-1} \nu_h^\pi \\ I_2 &:= \sum_{h_1 < h_2}^H (\nu_{h_1}^\pi)^\top \Sigma^{-1} \left(\frac{1}{K} \sum_{k=1}^K \sum_{h'=1}^{H_k} \varphi(s_{h'}^k, a_{h'}^k) \varphi(s_{h'}^k, a_{h'}^k)^\top \epsilon_{h_1 h'}^k \epsilon_{h_2 h'}^k \right) \Sigma^{-1} \nu_{h_2}^\pi \end{aligned}$$

Condition (38):

$$\sum_{k=1}^K \sum_{h'=1}^{H_k} \left[\frac{1}{\sqrt{K}} \sum_{h=1}^H (\nu_h^\pi)^\top \Sigma^{-1} \varphi(s_{h'}^k, a_{h'}^k) \epsilon_{hh'}^k \right]^2 = I_1 + 2I_2$$

It's easy to verify that

$$I_1 = \sum_{h=1}^H (\nu_h^\pi)^\top \Sigma^{-\frac{1}{2}} \left(\frac{1}{K} \sum_{k=1}^K A_k \right) \Sigma^{-\frac{1}{2}} \nu_h^\pi, \text{ and } \bar{H}\sigma_1^2 = \sum_{h=1}^H (\nu_h^\pi)^\top \Sigma^{-\frac{1}{2}} \mathbb{E}(A_k) \Sigma^{-\frac{1}{2}} \nu_h^\pi,$$

where we denote

$$A_k := \sum_{h'=1}^{H_k} B_{hh'}^k := \sum_{h'=1}^{H_k} \Sigma^{-\frac{1}{2}} \varphi(s_{h'}^k, a_{h'}^k) \varphi(s_{h'}^k, a_{h'}^k)^\top (\epsilon_{hh'}^k)^2 \Sigma^{-\frac{1}{2}}$$

Thus, we have

$$\begin{aligned} \left| I_1 - \bar{H}\sigma_1^2 \right| &= \left| \sum_{h=1}^H (\nu_h^\pi)^\top \Sigma^{-\frac{1}{2}} \left(\frac{1}{K} \sum_{k=1}^K A_k - \mathbb{E}(A_k) \right) \Sigma^{-\frac{1}{2}} \nu_h^\pi \right| \\ &\leq \sum_{h=1}^H \left\| (\nu_h^\pi)^\top \Sigma^{-\frac{1}{2}} \right\|_2^2 \left\| \frac{1}{K} \sum_{k=1}^K (A_k - \mathbb{E}(A_k)) \right\|_2 \end{aligned}$$

Let $S_k = A_k - \mathbb{E}(A_k)$, and the (i, j) th entry of $S_k, A_k, B_{hh'}^k$ to be $s_{ij}^{(k)}, a_{ij}^{(k)}, b_{h',i,j}^{(k)}$ respectively.

$$\begin{aligned} \left\| B_{hh'}^k \right\|_2 &= \left\| \Sigma^{-\frac{1}{2}} \varphi(s_{h'}^k, a_{h'}^k) \varphi(s_{h'}^k, a_{h'}^k)^\top (\epsilon_{hh'}^k)^2 \Sigma^{-\frac{1}{2}} \right\|_2 \\ &\leq (H - h + 1)^2 \left\| \Sigma^{-\frac{1}{2}} \varphi(s_{h'}^k, a_{h'}^k) \right\|_2^2 \\ &= (H - h + 1)^2 \varphi(s_{h'}^k, a_{h'}^k)^\top \Sigma^{-1} \varphi(s_{h'}^k, a_{h'}^k) \\ &\leq C_1 d (H - h + 1)^2 \\ |b_{h',i,j}^{(k)}| &\leq \left\| B_{hh'}^k \right\|_2 \leq C_1 d (H - h + 1)^2 \end{aligned}$$

By Lemma A.5, we have that $s_{ij}^{(k)} = a_{ij}^{(k)} - \mathbb{E}(a_{ij}^{(k)})$ is sub-exponential with the sub-exponential norm at most

$$\|s_{ij}^{(k)}\|_{\psi_1} = C_1 d (H - h + 1)^2 \left(\|H_k\|_{\psi_1} + \frac{1}{\ln 2} \bar{H} \right) \quad (40)$$

Since $\mathcal{D}_k$ s are i.i.d, by the Bernstein inequality for sub-exponential random variables in Vershynin (2019), we have

$$\mathbb{P} \left(\left| \sum_{k=1}^K s_{ij}^{(k)} \right| \geq \epsilon \right) \leq 2 \exp \left(-c \min \left(\frac{\epsilon^2}{K \|s_{ij}^{(k)}\|_{\psi_1}^2}, \frac{\epsilon}{\|s_{ij}^{(k)}\|_{\psi_1}} \right) \right), \forall \epsilon \geq 0.$$

where $c > 0$ is an absolute value. With probability at least $1 - \delta$, and for $K \geq \frac{1}{c} \log(2/\delta)$,

$$\left| \frac{1}{K} \sum_{k=1}^K s_{ij}^{(k)} \right| \leq \frac{1}{\sqrt{K}} \sqrt{c \log(2/\delta)} \|s_{ij}^{(k)}\|_{\psi_1}$$

Therefore,

$$\frac{1}{K} \left| \sum_{k=1}^K \left(a_{ij}^{(k)} - \mathbb{E} \left(a_{ij}^{(k)} \right) \right) \right| \xrightarrow{p} 0,$$

uniformly for all $i, j = 1, \dots, d$.

Consequently,

$$\begin{aligned} \frac{1}{K} \left\| \sum_{k=1}^K (A_k - \mathbb{E}(A_k)) \right\|_2 &\leq \frac{1}{K} \left\| \sum_{k=1}^K (A_k - \mathbb{E}(A_k)) \right\|_F \\ &\leq \frac{d}{K} \max_{i,j} \left| \sum_{k=1}^K \left(a_{ij}^{(k)} - \mathbb{E} \left(a_{ij}^{(k)} \right) \right) \right| \xrightarrow{p} 0 \end{aligned}$$

as $K \rightarrow \infty$.

Finally, we obtain $I_1 \xrightarrow{p} \bar{H}\sigma_1^2$ as $K \rightarrow \infty$.

For I_2 , notice that

$$\left| I_2 - \bar{H}\sigma_2^2 \right| \leq \sum_{h_1 < h_2}^H \left\| (\nu_{h_1}^\pi)^\top \Sigma^{-\frac{1}{2}} \right\|_2 \left\| (\nu_{h_2}^\pi)^\top \Sigma^{-\frac{1}{2}} \right\|_2 \left\| \frac{1}{K} \sum_{k=1}^K (A_{k,h_1,h_2} - \mathbb{E}(A_{k,h_1,h_2})) \right\|_2$$

where

$$A_{k,h_1,h_2} := \sum_{h'=1}^{H_k} B_{h_1,h_2,h'}^k := \sum_{h'=1}^{H_k} \Sigma^{-\frac{1}{2}} \varphi(s_{h'}^k, a_{h'}^k) \varphi(s_{h'}^k, a_{h'}^k)^\top \epsilon_{h_1 h'}^k \epsilon_{h_2 h'}^k \Sigma^{-\frac{1}{2}}$$

Following almost the same proof as shown for I_1 , we obtain $I_2 \xrightarrow{p} \bar{H}\sigma_2^2$ as $K \rightarrow \infty$. Putting everything together, condition (38) has been verified. On the other hand, $1/\sqrt{\bar{H}_k} \xrightarrow{p} 1/\sqrt{\bar{H}}$. Applying the martingale central limit theorem and Slutsky's theorem, we obtain that $E_1 \xrightarrow{d} \mathcal{N}(0, \sigma_1^2 + 2\sigma_2^2)$. $\blacksquare$

Meanwhile, we have the following lemma:

Lemma A.3. *For*

$$\Sigma_0 = \frac{1}{\bar{H}} \mathbb{E} \left[\sum_{h'=1}^{H_1} \varphi(s_{h'}^1, a_{h'}^1) \varphi(s_{h'}^1, a_{h'}^1)^\top \right], \quad (41)$$

$$\Sigma = \mathbb{E} \left[\frac{1}{N} \sum_{k=1}^K \sum_{h'=1}^{H_k} \varphi(s_{h'}^k, a_{h'}^k) \varphi(s_{h'}^k, a_{h'}^k)^\top \right], \quad (42)$$

we have

$$\Sigma \rightarrow \Sigma_0 \quad \text{as } K \rightarrow \infty.$$

Based on this result, we have the following corollary:

Corollary A.4. $E_1 \xrightarrow{d} \mathcal{N}(0, \sigma^2)$, where

$$\sigma^2 = \frac{1}{\bar{H}} \sum_{h=1}^H (\nu_h^\pi)^\top \Sigma_0^{-1} \Omega_{h,h} \Sigma_0^{-1} \nu_h^\pi + \frac{2}{\bar{H}} \sum_{h_1 < h_2}^H (\nu_{h_1}^\pi)^\top \Sigma_0^{-1} \Omega_{h_1,h_2} \Sigma_0^{-1} \nu_{h_2}^\pi. \quad (43)$$

Lemma A.5. (Random variables upperbounded by sub-Exponential r.v. are sub-Exponential)

For random variables X , $|X| \leq C_b Y$, $C_b \geq 0$, where Y is positive sub-exponential random variable with sub-exponential norm $\|Y\|_{\psi_1} = M$, $Z = X - \mathbb{E}(X)$ is sub-exponential with the sub-exponential norm at most

$$\|Z\|_{\psi_1} \leq C_b \left(M + \frac{1}{\ln 2} \mathbb{E}(Y) \right) \quad (44)$$

A.3 E_2

Recall that

$$E_2 = \sum_{h=1}^H \left(N (\hat{\nu}_h^\pi)^\top \hat{\Sigma}^{-1} - (\nu_h^\pi)^\top \Sigma^{-1} \right) \left(\frac{1}{\sqrt{N}} \sum_{k=1}^K \sum_{h'=1}^{H_k} \varphi(s_{h'}^k, a_{h'}^k) (Q_h^\pi(s_{h'}^k, a_{h'}^k) - r_{h'}^k - V_{h+1}^\pi(s_{h'+1}^k)) \right)$$

Theorem A.6 (High-Order Term E_2). Under the Assumption 3, $|E_2| \xrightarrow{p} 0$.

In order to prove A.6, we first decompose E_2 into terms that are tractable to control.

Lemma A.7 (Decomposition of E_2). • It always holds that

$$|E_2| \leq \sqrt{N} \sum_{h=1}^H \sqrt{(\nu_1^\pi)^\top (\Sigma^\pi)^{-1} \nu_1^\pi} \left\| (\Sigma^\pi)^{-\frac{1}{2}} \Sigma^{-\frac{1}{2}} \right\|_2 \left\| \Sigma^{-\frac{1}{2}} \Delta W_h^\pi \right\|_2 \quad (45)$$

$$\left[\left(1 + \left\| (\Sigma^\pi)^{\frac{1}{2}} \Delta M^\pi (\Sigma^\pi)^{-\frac{1}{2}} \right\|_2 \right)^{h-1} \left(1 + \left\| \Sigma^{\frac{1}{2}} \Delta X \Sigma^{\frac{1}{2}} \right\|_2 \right) - 1 \right]$$

where

$$\Delta W_h^\pi = \frac{1}{N} \sum_{k=1}^K \sum_{h'=1}^{H_k} \varphi(s_{h'}^k, a_{h'}^k) (Q_h^\pi(s_{h'}^k, a_{h'}^k) - r_{h'}^k - V_{h+1}^\pi(s_{h'+1}^k)) \quad (46)$$

•

$$\left\| (\Sigma^\pi)^{\frac{1}{2}} \Delta M^\pi (\Sigma^\pi)^{-\frac{1}{2}} \right\|_2 \leq \sqrt{\kappa_1} \left[\left(1 + \left\| \Sigma^{\frac{1}{2}} \Delta X \Sigma^{\frac{1}{2}} \right\|_2 \right) \left(1 + \left\| \Sigma^{-\frac{1}{2}} \Delta Y^\pi \Sigma^{-\frac{1}{2}} \right\|_2 \right) - 1 \right] \quad (47)$$

where

$$\Delta Y^\pi = \frac{1}{N} \sum_{k=1}^K \sum_{h'=1}^{H_k} \varphi(s_{h'}^k, a_{h'}^k) \varphi^\pi(s_{h'+1}^k)^\top - \Sigma M^\pi \quad (48)$$

$$\kappa_1 = \text{cond} \left(\Sigma^{-\frac{1}{2}} \Sigma^\pi \Sigma^{-\frac{1}{2}} \right) \quad (49)$$

• If $\|N^{-1} \Sigma^{-\frac{1}{2}} \hat{\Sigma} \Sigma^{-\frac{1}{2}} - I\|_2 \leq \frac{1}{2}$, then

$$\left\| \Sigma^{\frac{1}{2}} \Delta X \Sigma^{\frac{1}{2}} \right\|_2 \leq 2 \left\| N^{-1} \Sigma^{-\frac{1}{2}} \hat{\Sigma} \Sigma^{-\frac{1}{2}} - I \right\|_2 \quad (50)$$

Proof. The proof is the same as Lemma B.7 in Duan and Wang (2020). ■

Lemma A.7 shows that the problem is now reduced to estimating $\|N^{-1}\Sigma^{-\frac{1}{2}}\hat{\Sigma}\Sigma^{-\frac{1}{2}} - I\|_2$, $\|\Sigma^{-\frac{1}{2}}\Delta Y^\pi\Sigma^{-\frac{1}{2}}\|_2$, $\|\Sigma^{-\frac{1}{2}}\Delta W_h^\pi\|_2$.

Lemma A.8. *Under the Assumption 3, with probability at least $1 - \delta$, for $K \geq \frac{1}{c} \log(4/\delta)$*

$$\left\| N^{-1}\Sigma^{-\frac{1}{2}}\hat{\Sigma}\Sigma^{-\frac{1}{2}} - I \right\|_2 \leq \frac{Cd^2}{\bar{H}} \sqrt{\frac{\log 4d^2/\delta}{cK}} \|H_k\|_{\psi_1} + \frac{\lambda}{K} \|\Sigma^{-1}\|_2 \quad (51)$$

Corollary A.9. *Under the Assumption 3, $\left\| N^{-1}\Sigma^{-\frac{1}{2}}\hat{\Sigma}\Sigma^{-\frac{1}{2}} - I \right\|_2 \xrightarrow{p} 0$*

Lemma A.10. *Under the Assumption 3, with probability at least $1 - \delta$, for $K \geq \frac{1}{c} \log(4/\delta)$*

$$\left\| \Sigma^{-\frac{1}{2}}\Delta Y^\pi\Sigma^{-\frac{1}{2}} \right\|_2 \leq \frac{Cd^2}{\bar{H}} \sqrt{\frac{\log 4d^2/\delta}{cK}} \|H_k\|_{\psi_1} \quad (52)$$

Consequently,

$$\left\| \Sigma^{-\frac{1}{2}}\Delta Y^\pi\Sigma^{-\frac{1}{2}} \right\|_2 \xrightarrow{p} 0. \quad (53)$$

Lemma A.11. *Under the Assumption 3, with probability at least $1 - \delta$, for $K \geq \frac{1}{c} \log(4/\delta)$*

$$\left\| \Sigma^{-\frac{1}{2}}\Delta W_h^\pi \right\|_2 \leq \sqrt{\frac{d^3 \log(4d^2/\delta)}{cK}} \frac{\|H_k\|_{\varphi_1}}{\bar{H}} (H - h + 1) \quad (54)$$

We now prove A.6.

Proof of A.6. From Corollary A.9 we have known that,

$$\left\| N^{-1}\Sigma^{-\frac{1}{2}}\hat{\Sigma}\Sigma^{-\frac{1}{2}} - I \right\|_2 \xrightarrow{p} 0 \quad (55)$$

Plugging it into (50), by Slutsky's theorem, we obtain,

$$\left\| \Sigma^{\frac{1}{2}}\Delta X\Sigma^{\frac{1}{2}} \right\|_2 \xrightarrow{p} 0 \quad (56)$$

Furthermore, plugging them into (47), by Slutsky's theorem,

$$\left\| (\Sigma^\pi)^{\frac{1}{2}} \Delta M^\pi (\Sigma^\pi)^{-\frac{1}{2}} \right\|_2 \xrightarrow{p} 0 \quad (57)$$

Then, for each h ,

$$\left[\left(1 + \left\| (\Sigma^\pi)^{\frac{1}{2}} \Delta M^\pi (\Sigma^\pi)^{-\frac{1}{2}} \right\|_2 \right)^h \left(1 + \left\| \Sigma^{\frac{1}{2}}\Delta X\Sigma^{\frac{1}{2}} \right\|_2 \right) - 1 \right] \xrightarrow{p} 0 \quad (58)$$

From Lemma A.11, we know that with probability at least $1 - \delta$, for $K \geq \frac{1}{c} \log(4/\delta)$

$$\sqrt{K} \left\| \Sigma^{-\frac{1}{2}}\Delta W_h^\pi \right\|_2 \leq \sqrt{d^3 \log(4d^2/\delta)} \frac{\|H_k\|_{\varphi_1}}{\bar{H}} H \quad (59)$$

That is, $\sqrt{K} \left\| \Sigma^{-\frac{1}{2}}\Delta W_h^\pi \right\|_2$ is bounded in probability. Combining (58) (59), and that $\sqrt{\bar{H}_k} \xrightarrow{p} \sqrt{\bar{H}}$, we yield $E_2 \xrightarrow{p} 0$. ■

A.4 The Bias Term E_3

Theorem A.12 (The Bias Term E_3). *For bounded λ , $|E_3| \xrightarrow{p} 0$.*

Proof. Recall

$$\begin{aligned}
E_3 &= \sqrt{N} \sum_{h=1}^H \lambda (\hat{\nu}_h^\pi)^T \hat{\Sigma}^{-1} w_h \\
&= \sqrt{N} \lambda \sum_{h=1}^H (\hat{\nu}_1^\pi)^T \left(\widehat{M} \right)^{h-1} \hat{\Sigma}^{-1} w_h \\
&= \frac{\lambda}{\sqrt{N}} \sum_{h=1}^H \left((\hat{\nu}_1^\pi)^T (\Sigma^\pi)^{-\frac{1}{2}} \right) \left((\Sigma^\pi)^{\frac{1}{2}} \widehat{M} (\Sigma^\pi)^{-\frac{1}{2}} \right)^{h-1} \left((\Sigma^\pi)^{\frac{1}{2}} \Sigma^{-\frac{1}{2}} \right) \\
&\quad \cdot \left(N \Sigma^{\frac{1}{2}} \hat{\Sigma}^{-1} \Sigma^{\frac{1}{2}} \right) \Sigma^{-1} \left(\Sigma^{\frac{1}{2}} w_h \right).
\end{aligned}$$

Thus,

$$\begin{aligned}
|E_3| &\leq \frac{\lambda}{\sqrt{N}} \sum_{h=1}^H \sqrt{(\hat{\nu}_1^\pi)^T (\Sigma^\pi)^{-1} \hat{\nu}_1^\pi} \left\| (\Sigma^\pi)^{\frac{1}{2}} \widehat{M} (\Sigma^\pi)^{-\frac{1}{2}} \right\|_2^{h-1} \left\| (\Sigma^\pi)^{\frac{1}{2}} \Sigma^{-\frac{1}{2}} \right\|_2 \\
&\quad \times \left\| N \Sigma^{\frac{1}{2}} \hat{\Sigma}^{-1} \Sigma^{\frac{1}{2}} \right\|_2 \left\| \Sigma^{-1} \right\|_2 \left\| \Sigma^{\frac{1}{2}} w_h \right\|_2
\end{aligned} \tag{60}$$

From (57) and Lemma A.7, we have that

$$\left\| (\Sigma^\pi)^{\frac{1}{2}} \widehat{M} (\Sigma^\pi)^{-\frac{1}{2}} \right\|_2 \leq \left\| (\Sigma^\pi)^{\frac{1}{2}} M (\Sigma^\pi)^{-\frac{1}{2}} \right\|_2 + \left\| (\Sigma^\pi)^{\frac{1}{2}} \Delta M (\Sigma^\pi)^{-\frac{1}{2}} \right\|_2 \leq 1 + \left\| (\Sigma^\pi)^{\frac{1}{2}} \Delta M (\Sigma^\pi)^{-\frac{1}{2}} \right\|_2$$

The second inequality is by the contraction of Markov chain, see Lemma B.1 in [Duan and Wang \(2020\)](#). Then, $\left\| (\Sigma^\pi)^{\frac{1}{2}} \widehat{M} (\Sigma^\pi)^{-\frac{1}{2}} \right\|_2$ is bounded in probability.

Likewise, from (56), it follows that

$$\left\| N \Sigma^{\frac{1}{2}} \hat{\Sigma}^{-1} \Sigma^{\frac{1}{2}} \right\|_2 = \left\| \Sigma^{\frac{1}{2}} (\Sigma^{-1} + \Delta X) \Sigma^{\frac{1}{2}} \right\|_2 = \left\| I + \Sigma^{\frac{1}{2}} \Delta X \Sigma^{\frac{1}{2}} \right\|_2 \leq 1 + \left\| \Sigma^{\frac{1}{2}} \Delta X \Sigma^{\frac{1}{2}} \right\|_2$$

bounded in probability.

The rest of the terms on the right-hand side of (60) are deterministic, and we also know that

$$\begin{aligned}
\left\| \Sigma^{\frac{1}{2}} w_h \right\|_2^2 &= W_n^\top \Sigma W_n = \mathbb{E} \left[\frac{1}{N} \sum_{k=1}^K \sum_{h'=1}^{H_k} w_h^\top \varphi(s_{h'}^k, a_{h'}^k) \varphi(s_{h'}^k, a_{h'}^k)^\top w_h \right] \\
&= \mathbb{E} \left[\frac{1}{N} \sum_{k=1}^K \sum_{h'=1}^{H_k} \mathbb{E} \left[\left(\varphi(s_{h'}^k, a_{h'}^k)^\top w_h \right)^2 \middle| H_1, \dots, H_K \right] \right] \\
&\leq \mathbb{E} \left[\frac{1}{N} \sum_{k=1}^K \sum_{h'=1}^{H_k} (H - h + 1)^2 \right] = (H - h + 1)^2
\end{aligned}$$

Meanwhile, $1/\sqrt{N} \xrightarrow{p} 0$. Hence, for a bounded λ , $|E_3| \xrightarrow{p} 0$. ■

B Proofs of Lemmas in Appendix A

B.1 Proof of A.3

Proof. Recall that

$$\begin{aligned}\Sigma_0 &= \frac{1}{H} \mathbb{E} \left[\sum_{h'=1}^{H_1} \varphi(s_{h'}^1, a_{h'}^1) \varphi(s_{h'}^1, a_{h'}^1)^\top \right], \\ \Sigma &= \mathbb{E} \left[\frac{1}{N} \sum_{k=1}^K \sum_{h'=1}^{H_k} \varphi(s_{h'}^k, a_{h'}^k) \varphi(s_{h'}^k, a_{h'}^k)^\top \right],\end{aligned}$$

By the law of large number, we have

$$\frac{1}{K} \sum_{k=1}^K \sum_{h'=1}^{H_k} \varphi(s_{h'}^k, a_{h'}^k) \varphi(s_{h'}^k, a_{h'}^k)^\top \xrightarrow{p} \mathbb{E} \left[\sum_{h'=1}^{H_1} \varphi(s_{h'}^1, a_{h'}^1) \varphi(s_{h'}^1, a_{h'}^1)^\top \right]$$

and $1/\bar{H}_k \xrightarrow{p} 1/\bar{H}$. Then

$$\frac{1}{N} \sum_{k=1}^K \sum_{h'=1}^{H_k} \varphi(s_{h'}^k, a_{h'}^k) \varphi(s_{h'}^k, a_{h'}^k)^\top \xrightarrow{p} \Sigma_0.$$

Meanwhile, by Assumption 3, the (i, j) th entry is bounded, i.e.

$$\left| \left(\frac{1}{N} \sum_{k=1}^K \sum_{h'=1}^{H_k} \varphi(s_{h'}^k, a_{h'}^k) \varphi(s_{h'}^k, a_{h'}^k)^\top \right)_{i,j} \right| \leq 1.$$

so it is also uniformly integrable. By the Vitali Convergence Theorem,

$$\left(\frac{1}{N} \sum_{k=1}^K \sum_{h'=1}^{H_k} \varphi(s_{h'}^k, a_{h'}^k) \varphi(s_{h'}^k, a_{h'}^k)^\top \right)_{i,j} \xrightarrow{L_1} (\Sigma_0)_{i,j}$$

then

$$\Sigma = \mathbb{E} \left[\frac{1}{N} \sum_{k=1}^K \sum_{h'=1}^{H_k} \varphi(s_{h'}^k, a_{h'}^k) \varphi(s_{h'}^k, a_{h'}^k)^\top \right] \rightarrow \Sigma_0.$$

■

B.2 Proof of A.5

Proof. Since Y is sub-exponential with norm $\|Y\|_{\psi_1} = M$,

$$\mathbb{E}(\exp(t|X|)) \leq \mathbb{E}(e^{tC_b Y}) \leq e^{M^2 C_b t}, \forall t \in \left(0, \frac{1}{M^2 C_b}\right).$$

By the definition of the sub-exponential random variable in Vershynin (2019), we know that X is sub-exponential with $\|X\|_{\psi_1}$ at most MC_b .

Meanwhile, $\mathbb{E}(X)$ is a constant and

$$|\mathbb{E}(X)| \leq \mathbb{E}(|X|) \leq C_b \mathbb{E}(Y)$$

Computing the sub-exponential norm of $\mathbb{E}(X)$ yields,

$$\|\mathbb{E}(X)\|_{\psi_1} \leq \|\mathbb{E}(X)\|_{\psi_2} \|1\|_{\psi_2} = \frac{1}{\sqrt{\ln 2}} \|\mathbb{E}(X)\|_{\psi_2} \leq \frac{1}{\ln 2} \mathbb{E}(|X|) \leq \frac{C_b}{\ln 2} \mathbb{E}(Y)$$

The second inequality is due to the property of sub-Gaussian r.v., see section 2.5.2 in [Vershynin \(2019\)](#). Therefore,

$$\|Z\|_{\psi_1} \leq \|X\|_{\psi_1} + \|\mathbb{E}(X)\|_{\psi_1} \leq C_b \left(M + \frac{1}{\ln 2} \mathbb{E}(Y) \right)$$

■

B.3 Proof of [A.8](#)

Proof. Let

$$Z_K = \frac{1}{N} \sum_{k=1}^K \sum_{h'=1}^{H_k} \Sigma^{-\frac{1}{2}} \varphi(s_{h'}^k, a_{h'}^k) \varphi(s_{h'}^k, a_{h'}^k)^\top \Sigma^{-\frac{1}{2}},$$

so we could write

$$N^{-1} \Sigma^{-\frac{1}{2}} \hat{\Sigma} \Sigma^{-\frac{1}{2}} - I = Z_K - I + \frac{\lambda}{N} \Sigma^{-1}$$

Since $H_k \geq 1$, $\frac{\lambda}{N} \|\Sigma^{-1}\|_2 \leq \frac{\lambda}{K} \|\Sigma^{-1}\|_2$.

Let

$$A_k := \sum_{h'=1}^{H_k} B_{hh'}^k := \sum_{h'=1}^{H_k} \Sigma^{-\frac{1}{2}} \varphi(s_{h'}^k, a_{h'}^k) \varphi(s_{h'}^k, a_{h'}^k)^\top \Sigma^{-\frac{1}{2}},$$

then $Z_K = \sum_{k=1}^K A_k / \sum_{k=1}^K H_k$, and notice that $E(Z_K) = I$, we further denote that $\tilde{Z}_K = \frac{1}{K\bar{H}} \sum_{k=1}^K A_k$.

Meanwhile,

$$\begin{aligned} \left\| B_{hh'}^k \right\|_2 &= \left\| \Sigma^{-\frac{1}{2}} \varphi(s_{h'}^k, a_{h'}^k) \varphi(s_{h'}^k, a_{h'}^k)^\top (\epsilon_{hh'}^k)^2 \Sigma^{-\frac{1}{2}} \right\|_2 \\ &\leq \left\| \Sigma^{-\frac{1}{2}} \varphi(s_{h'}^k, a_{h'}^k) \right\|_2^2 \\ &= \varphi(s_{h'}^k, a_{h'}^k)^\top \Sigma^{-1} \varphi(s_{h'}^k, a_{h'}^k) \\ &\leq C_1 d \end{aligned} \tag{61}$$

so $\|A_k\|_2 \leq C_1 d H_k$, and that

$$\begin{aligned} \mathbb{P} \left(\|Z_K - \tilde{Z}_K\| \geq \epsilon_{11} \right) &\leq \mathbb{P} \left(\left| \frac{1}{K} \left| \frac{1}{\bar{H}_k} - \frac{1}{\bar{H}} \right| \sum_{k=1}^K \|A_k\|_2 \geq \epsilon_{11} \right) \\ &\leq \mathbb{P} \left(\left| \frac{1}{\bar{H}_k} - \frac{1}{\bar{H}} \right| \frac{1}{K} \sum_{k=1}^K C_1 d H_k \geq \epsilon_{11} \right) \\ &= \mathbb{P} \left(\frac{C_1 d}{\bar{H}} \left| \bar{H}_k - \bar{H} \right| \geq \epsilon_{11} \right) \\ &\leq 2 \exp \left(-c \frac{\epsilon_{11}^2 \bar{H}^2 K}{C_1 d^2 (\|H_k\|_{\psi_1} + \bar{H} / \ln 2)^2} \right), \end{aligned}$$

if $\epsilon_{11} \in [0, (\|H_k\|_{\psi_1} + \frac{1}{\ln 2} \bar{H}) C_1 d / \bar{H}]$. The last inequality is the Bernstein inequality from [Vershynin \(2019\)](#), Corollary 2.8.3. Thus, for $K \geq \frac{1}{c} \log(4/\delta)$,

$$\mathbb{P}\left(\|Z_K - \tilde{Z}_K\| \geq \epsilon_{11}\right) \leq \frac{\delta}{2}, \text{ where } \epsilon_{11} = \frac{C_1 d}{\bar{H}} \left(\|H_k\|_{\psi_1} + \frac{1}{\ln 2} \bar{H}\right) \sqrt{\frac{\log 4/\delta}{cK}},$$

Let $S_k = A_k - \mathbb{E}(A_k)$, and the (i, j) th entry of $S_k, A_k, B_{hh'}^k$ to be $s_{ij}^{(k)}, a_{ij}^{(k)}, b_{h',i,j}^{(k)}$ respectively. Then,

$$|b_{h',i,j}^{(k)}| \leq \left\| B_{hh'}^k \right\|_2 \leq C_1 d \quad (62)$$

By Lemma A.5, we have that $s_{ij}^{(k)} = a_{ij}^{(k)} - \mathbb{E}(a_{ij}^{(k)})$ is sub-exponential with the sub-exponential norm, denoted as $M_s := \|s_{ij}^{(k)}\|_{\psi_1}$, at most

$$M_s = C_1 d \left(\|H_k\|_{\psi_1} + \frac{1}{\ln 2} \bar{H}\right) \quad (63)$$

Since $\mathcal{D}_k$ s are i.i.d, by the Bernstein inequality for sub-exponential random variables in [Vershynin \(2019\)](#), we have

$$\mathbb{P}\left(\left|\sum_{k=1}^K s_{ij}^{(k)}\right| \geq \epsilon\right) \leq 2 \exp\left(-c \min\left(\frac{\epsilon^2}{K \|s_{ij}^{(k)}\|_{\psi_1}^2}, \frac{\epsilon}{\|s_{ij}^{(k)}\|_{\psi_1}}\right)\right), \forall \epsilon \geq 0.$$

where $c > 0$ is an absolute value. For $K \geq \frac{1}{c} \log(2/\delta)$,

$$\mathbb{P}\left(\left|\frac{1}{K} \sum_{k=1}^K s_{ij}^{(k)}\right| \geq \sqrt{\frac{\log(2/\delta)}{cK}} \|s_{ij}^{(k)}\|_{\psi_1}\right) \leq \delta \quad (64)$$

uniformly for all $i, j = 1, \dots, d$.

Notice that

$$\frac{1}{K \bar{H}} \left\| \sum_{k=1}^K (A_k - \mathbb{E}(A_k)) \right\|_2 \leq \frac{1}{K \bar{H}} \left\| \sum_{k=1}^K (A_k - \mathbb{E}(A_k)) \right\|_F \leq \frac{d}{K \bar{H}} \max_{i,j} \left| \sum_{k=1}^K s_{ij}^{(k)} \right|$$

Then for $K \geq \frac{1}{c} \log(4/\delta)$,

$$\begin{aligned} \mathbb{P}\left(\|\tilde{Z}_K - \mathbb{E}(\tilde{Z}_K)\| \geq \epsilon_{12}\right) &= \mathbb{P}\left(\frac{1}{K \bar{H}} \left\| \sum_{k=1}^K (A_k - \mathbb{E}(A_k)) \right\|_2 \geq \epsilon_{12}\right) \\ &\leq \mathbb{P}\left(\frac{d}{K \bar{H}} \max_{i,j} \left| \sum_{k=1}^K s_{ij}^{(k)} \right| \geq \epsilon_{12}\right) \\ &\leq \mathbb{P}\left(\max_{i,j} \left| \frac{1}{K} \sum_{k=1}^K s_{ij}^{(k)} \right| \geq \frac{\bar{H} \epsilon_{12}}{d}\right) \\ &\leq \sum_{i,j} \mathbb{P}\left(\left| \frac{1}{K} \sum_{k=1}^K s_{ij}^{(k)} \right| \geq \frac{\bar{H} \epsilon_{12}}{d}\right) \\ &\leq \frac{\delta}{2}, \end{aligned}$$

The last inequality is because by (64), we have

$$\sup_{i,j} \mathbb{P} \left(\left| \frac{1}{K} \sum_{k=1}^K s_{ij}^{(k)} \right| \geq \sqrt{\frac{\log(4d^2/\delta)}{cK}} M_s \right) \leq \frac{\delta}{2d^2},$$

then let

$$\epsilon_{12} = \frac{d}{\bar{H}} \sqrt{\frac{\log(4d^2/\delta)}{cK}} M_s = \frac{C_1 d^2}{\bar{H}} \sqrt{\frac{\log 4d^2/\delta}{cK}} \left(\|H_k\|_{\psi_1} + \frac{1}{\ln 2} \bar{H} \right).$$

On the other hand,

$$\|\mathbb{E}(\tilde{Z}_K) - \mathbb{E}(Z_K)\| \leq \frac{1}{K} \sum_{k=1}^K \mathbb{E} \left\| \left(\frac{1}{\bar{H}_k} - \frac{1}{\bar{H}} \right) A_k \right\| \leq \frac{C_1 d}{\bar{H}} \mathbb{E} |\bar{H}_k - \bar{H}| \leq \frac{Cd \|H_k\|_{\psi_1}}{\bar{H} \sqrt{K}}$$

The last inequality is by ... Consequently,

$$\begin{aligned} & \mathbb{P} \left(\|Z_K - I\| \geq \epsilon_{11} + \epsilon_{12} + \frac{Cd \|H_k\|_{\psi_1}}{\bar{H} \sqrt{K}} \right) \\ & \leq \mathbb{P} \left(\|Z_K - I\| \geq \epsilon_{11} + \epsilon_{12} + \|\mathbb{E}(\tilde{Z}_K) - \mathbb{E}(Z_K)\| \right) \\ & \leq \mathbb{P} \left(\|Z_K - \tilde{Z}_K\| + \|\tilde{Z}_K - \mathbb{E}(\tilde{Z}_K)\| + \|\mathbb{E}(\tilde{Z}_K) - \mathbb{E}(Z_K)\| \geq \epsilon_{11} + \epsilon_{12} + \|\mathbb{E}(\tilde{Z}_K) - \mathbb{E}(Z_K)\| \right) \\ & \leq \mathbb{P} \left(\|Z_K - \tilde{Z}_K\| \geq \epsilon_{11} \right) + \mathbb{P} \left(\|\tilde{Z}_K - \mathbb{E}(\tilde{Z}_K)\| \geq \epsilon_{12} \right) \\ & \leq \frac{\delta}{2} + \frac{\delta}{2} = \delta, \end{aligned}$$

Thus, for $K \geq \frac{1}{c} \log(4/\delta)$, with probability at least $1 - \delta$,

$$\begin{aligned} \|Z_K - I\| & \leq \epsilon_{11} + \epsilon_{12} + \frac{Cd \|H_k\|_{\psi_1}}{\bar{H} \sqrt{K}} \\ & = \frac{C_1 d^2}{\bar{H}} \sqrt{\frac{\log 4d^2/\delta}{cK}} \left(\|H_k\|_{\psi_1} + \frac{1}{\ln 2} \bar{H} \right) + \frac{Cd \|H_k\|_{\psi_1}}{\bar{H} \sqrt{K}} \\ & \leq \frac{Cd^2}{\bar{H}} \sqrt{\frac{\log 4d^2/\delta}{K}} \|H_k\|_{\psi_1} \end{aligned}$$

■

B.4 Proof of A.9

Proof. The only thing we have left to prove is $\mathbb{E} |\bar{H}_k - \bar{H}| \rightarrow 0$ as $K \rightarrow \infty$. Notice that

$$\mathbb{E} |\bar{H}_k - \bar{H}| = \mathbb{E} (\bar{H}_k - \bar{H}) + 2\mathbb{E} (\bar{H}_k - \bar{H})^- = 2\mathbb{E} (\bar{H}_k - \bar{H})^-$$

where $(\bar{H}_k - \bar{H})^- = \max\{0, \bar{H} - \bar{H}_k\} \leq \bar{H}$. Then for any $\epsilon > 0$,

$$\begin{aligned}\mathbb{E}(\bar{H}_k - \bar{H})^- &= \int_{|\bar{H}_k - \bar{H}| > \epsilon} (\bar{H}_k - \bar{H})^- dP + \int_{|\bar{H}_k - \bar{H}| < \epsilon} (\bar{H}_k - \bar{H})^- dP \\ &\leq \int_{|\bar{H}_k - \bar{H}| > \epsilon} (\bar{H}_k - \bar{H})^- dP + \epsilon \\ &\leq \int_{|\bar{H}_k - \bar{H}| > \epsilon} \bar{H} dP + \epsilon \\ &= \bar{H} \mathbb{P}(|\bar{H}_k - \bar{H}| > \epsilon) + \epsilon \rightarrow 0\end{aligned}$$

since $\bar{H}_k \xrightarrow{P} \bar{H}$. ■

B.5 Proof of A.10

Proof. Let

$$Z_K^\pi = \frac{1}{N} \sum_{k=1}^K \sum_{h'=1}^{H_k} \Sigma^{-\frac{1}{2}} \varphi(s_{h'}^k, a_{h'}^k) \varphi^\pi(s_{h'+1}^k)^\top \Sigma^{-\frac{1}{2}}$$

and notice that

$$\begin{aligned}\mathbb{E}(Z_K^\pi) &= \Sigma^{-\frac{1}{2}} \mathbb{E} \left[\mathbb{E} \left(\frac{1}{N} \sum_{k=1}^K \sum_{h'=1}^{H_k} \varphi(s_{h'}^k, a_{h'}^k) \varphi^\pi(s_{h'+1}^k)^\top \middle| H_1, \dots, H_K \right) \right] \Sigma^{-\frac{1}{2}} \\ &= \Sigma^{-\frac{1}{2}} \mathbb{E} \left[\frac{1}{N} \sum_{k=1}^K \sum_{h'=1}^{H_k} \mathbb{E} \left(\varphi(s_{h'}^k, a_{h'}^k) \varphi^\pi(s_{h'+1}^k)^\top \middle| H_1, \dots, H_K \right) \right] \Sigma^{-\frac{1}{2}} \\ &= \Sigma^{-\frac{1}{2}} \mathbb{E} \left[\frac{1}{N} \sum_{k=1}^K \sum_{h'=1}^{H_k} \mathbb{E} \left(\varphi(s_{h'}^k, a_{h'}^k) \varphi^\pi(s_{h'+1}^k)^\top \right) \right] \Sigma^{-\frac{1}{2}} \\ &= \Sigma^{-\frac{1}{2}} \mathbb{E} \left[\frac{1}{N} \sum_{k=1}^K \sum_{h'=1}^{H_k} \mathbb{E} \left(\mathbb{E} \left(\varphi(s_{h'}^k, a_{h'}^k) \varphi^\pi(s_{h'+1}^k)^\top \middle| s_{h'}^k, a_{h'}^k \right) \right) \right] \Sigma^{-\frac{1}{2}} \\ &= \Sigma^{-\frac{1}{2}} \mathbb{E} \left[\frac{1}{N} \sum_{k=1}^K \sum_{h'=1}^{H_k} \mathbb{E} \left(\varphi(s_{h'}^k, a_{h'}^k) \mathbb{E} \left(\varphi^\pi(s_{h'+1}^k)^\top \middle| s_{h'}^k, a_{h'}^k \right) \right) \right] \Sigma^{-\frac{1}{2}} \\ &= \Sigma^{-\frac{1}{2}} \mathbb{E} \left[\frac{1}{N} \sum_{k=1}^K \sum_{h'=1}^{H_k} \mathbb{E} \left(\varphi(s_{h'}^k, a_{h'}^k) \varphi(s_{h'}^k, a_{h'}^k)^\top M^\pi \right) \right] \Sigma^{-\frac{1}{2}} \\ &= \Sigma^{-\frac{1}{2}} \Sigma M^\pi \Sigma^{-\frac{1}{2}} \\ &= \Sigma^{\frac{1}{2}} M^\pi \Sigma^{-\frac{1}{2}}\end{aligned}$$

Let

$$A_k^\pi = \sum_{h'=1}^{H_k} \Sigma^{-\frac{1}{2}} \varphi(s_{h'}^k, a_{h'}^k) \varphi^\pi(s_{h'+1}^k)^\top \Sigma^{-\frac{1}{2}},$$

then $Z_K^\pi = \sum_{k=1}^K A_k^\pi / \sum_{k=1}^K H_k$, and $\Sigma^{-\frac{1}{2}} \Delta Y^\pi \Sigma^{-\frac{1}{2}} = Z_K^\pi - \mathbb{E}(Z_K^\pi)$.

Meanwhile,

$$\begin{aligned} \left\| A_k^\pi \right\|_2 &\leq \sum_{h'=1}^{H_k} \left\| \Sigma^{-\frac{1}{2}} \varphi(s_{h'}^k, a_{h'}^k) \varphi^\pi(s_{h'+1}^k)^\top \Sigma^{-\frac{1}{2}} \right\|_2 \\ &\leq \sum_{h'=1}^{H_k} \left\| \Sigma^{-\frac{1}{2}} \varphi(s_{h'}^k, a_{h'}^k) \right\|_2 \left\| \varphi^\pi(s_{h'+1}^k)^\top \Sigma^{-\frac{1}{2}} \right\|_2 \leq C_1 d H_k \end{aligned}$$

Parallel to the proof in Lemma A.8, for $K \geq \frac{1}{c} \log(4/\delta)$, we reach our final result. $\blacksquare$

B.6 Proof of A.11

Proof. Recall the definition $\epsilon_{hh'}^k = Q_h^\pi(s_{h'}^k, a_{h'}^k) - r_{h'}^k - V_{h+1}^\pi(s_{h'+1}^k)$. Notice that

$$\begin{aligned} \mathbb{E} \left(\Sigma^{-\frac{1}{2}} \Delta W_h^\pi \right) &= \Sigma^{-\frac{1}{2}} \mathbb{E} \left[\mathbb{E} \left(\frac{1}{N} \sum_{k=1}^K \sum_{h'=1}^{H_k} \varphi(s_{h'}^k, a_{h'}^k) \epsilon_{hh'}^k \middle| H_1, \dots, H_K \right) \right] \\ &= \Sigma^{-\frac{1}{2}} \mathbb{E}_{H_1, \dots, H_K} \left[\frac{1}{N} \sum_{k=1}^K \sum_{h'=1}^{H_k} \mathbb{E} [\varphi(s_{h'}^k, a_{h'}^k) \epsilon_{hh'}^k] \right] \\ &= \Sigma^{-\frac{1}{2}} \mathbb{E}_{H_1, \dots, H_K} \left[\frac{1}{N} \sum_{k=1}^K \sum_{h'=1}^{H_k} \mathbb{E} [\mathbb{E} [\varphi(s_{h'}^k, a_{h'}^k) \epsilon_{hh'}^k | s_{h'}^k, a_{h'}^k]] \right] \\ &= \Sigma^{-\frac{1}{2}} \mathbb{E}_{H_1, \dots, H_K} \left[\frac{1}{N} \sum_{k=1}^K \sum_{h'=1}^{H_k} \mathbb{E} [\varphi(s_{h'}^k, a_{h'}^k) \mathbb{E} [\epsilon_{hh'}^k | s_{h'}^k, a_{h'}^k]] \right] \\ &= 0. \end{aligned} \tag{65}$$

Let

$$A_k^W = \sum_{h'=1}^{H_k} \Sigma^{-\frac{1}{2}} \varphi(s_{h'}^k, a_{h'}^k) \epsilon_{hh'}^k$$

and

$$\left\| A_k^W \right\|_2 \leq (H - h + 1) \sqrt{C_1 d} H_k$$

Also, similar to (65), we have

$$\mathbb{E} \left(\frac{1}{K \bar{H}} \sum_{k=1}^K A_k^W \right) = 0.$$

Parallel to the proof in Lemma A.8 and A.10, we have for $K \geq \frac{1}{c} \log(4/\delta)$,

$$\mathbb{P} \left(\left\| \Sigma^{-\frac{1}{2}} \Delta W_h^\pi - \frac{1}{K \bar{H}} \sum_{k=1}^K A_k^W \right\| \geq \epsilon_{11}^W \right) \leq \frac{\delta}{2}, \mathbb{P} \left(\left\| \frac{1}{K \bar{H}} \sum_{k=1}^K A_k^W \right\| \geq \epsilon_{12}^W \right) \leq \frac{\delta}{2},$$

where

$$\begin{aligned} \epsilon_{11}^W &= \left(\|H_k\|_{\psi_1} + \frac{1}{\ln 2} \bar{H} \right) \sqrt{\frac{\log 4d^2/\delta}{cK}} \frac{\sqrt{C_1 d} (H - h + 1)}{\bar{H}}, \\ \epsilon_{12}^W &= \left(\|H_k\|_{\psi_1} + \frac{1}{\ln 2} \bar{H} \right) \sqrt{\frac{\log 4d^2/\delta}{cK}} \frac{\sqrt{C_1 d^3} (H - h + 1)}{\bar{H}} \end{aligned}$$

Thus,

$$\begin{aligned}
\mathbb{P} \left(\left\| \Sigma^{-\frac{1}{2}} \Delta W_h^\pi \right\| \geq \epsilon_{11}^W + \epsilon_{12}^W \right) &\leq \mathbb{P} \left(\left\| \Sigma^{-\frac{1}{2}} \Delta W_h^\pi - \frac{1}{K\bar{H}} \sum_{k=1}^K A_k^W \right\| \geq \epsilon_{11}^W \right) \\
&\quad + \mathbb{P} \left(\left\| \frac{1}{K\bar{H}} \sum_{k=1}^K A_k^W \right\| \geq \epsilon_{12}^W \right) \\
&\leq \frac{\delta}{2} + \frac{\delta}{2} = \delta.
\end{aligned}$$

For $K \geq \frac{1}{c} \log(4/\delta)$, with probability at least $1 - \delta$,

$$\begin{aligned}
\left\| \Sigma^{-\frac{1}{2}} \Delta W_h^\pi \right\|_2 &\leq \epsilon_{11}^W + \epsilon_{12}^W \leq \sqrt{\frac{d^3 \log(4d^2/\delta)}{cK}} \frac{(H - h + 1) \|H_k\|_{\varphi_1}}{\bar{H}} \\
&\leq \sqrt{\frac{d^3 \log(4d^2/\delta)}{cK}} \frac{H \|H_k\|_{\varphi_1}}{\bar{H}}
\end{aligned}$$

■

References

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Supplementary Files (optional)